

strategy maximizes long-run growth and is called the Kelly rule. It is obtained by finding the portfolio that maximizes one-period log returns. Since this portfolio maximizes (very) long-run wealth, it is also called the *optimal growth portfolio*.

The Kelly optimal growth portfolio dominates all other portfolios with a sufficiently long time span.⁴¹ So for the very long-run investor, should we hold the optimal growth portfolio if it maximizes very long-run wealth? No. This was settled by Samuelson in the 1970s, but the question is raised periodically by the unconvinced.⁴² Samuelson wrote a cute paper in 1979, entirely written in words of one syllable, entitled “Why We Should Not Make Mean Log of Wealth Big though Years to Act Are Long,” to answer this question. (Not surprisingly, limiting yourself to words of one syllable makes a paper quite hard to read.)

In a one-period model, you can maximize the portfolio growth rate by holding the portfolio that maximizes expected log returns. But do you have a log utility function? Probably not. You trade off risk and return differently than a log investor and are better off holding a portfolio optimized for your own risk aversion and utility function. Over the long run, you will outperform by following the Kelly rule. But, there is risk in doing so, and you might not be able to tolerate this risk. Furthermore, the long run in the Kelly rule could be very, very long. And as Keynes famously observed, in the long run, we are all dead.

In summary, follow the Merton–Samuelson advice and not the Kelly rule. Find your optimal one-period portfolio holdings over broad asset classes or strategies. Rebalance back to fixed weights. This is optimal with i.i.d. returns, and it will earn you a rebalancing premium. If you can forecast returns well, you also have a long-run opportunistic portfolio available.

6. Stay the Course? Redux

Daniel has suffered large losses and is feeling skittish about sticking to her long-term plan. But she is fortunate that she has no immediate liabilities and her income, which covers her expenses, is relatively safe. Yet the losses seem to have changed her tolerance for risk. She has told her financial planner, Harrison, that her IPS needed a “total overhaul” and she could not afford such big losses going forward.

According to the long-run investment advice from Merton and Samuelson, Harrison should advise Daniel to rebalance. Rebalancing to fixed weights (or exposures) is optimal when returns are not predictable. Even though returns are

⁴¹ The formal mathematical statement is that there exists a number $M(W)$ that depends on current wealth W such that $\Pr(W_T \leq W)$ using the Kelly rule is strictly less than $\Pr(W_T \leq W)$ using any other portfolio for all $T > M(W)$. Thus as $T \rightarrow \infty$, the Kelly rule dominates any other rule.

⁴² See Samuelson (1971) and Merton and Samuelson (1974).